Boosting as a Regularized Path to a Maximum Margin Classifier

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Abstract

In this paper we study boosting methods from a new perspective. We build on recent work by Efron et al. to show that boosting approximately (and in some cases exactly) minimizes its loss criterion with an L_1 constraint. For the two most commonly used loss criteria (exponential and logistic log-likelihood), we further show that as the constraint diminishes, or equivalently as the boosting iterations proceed, the solution converges - in the separable case - to an "L_1-optimal" separating hyper-plane. This "L_1-optimal" separating hyper-plane has the property of maximizing the minimal margin of the training data, as defined in the boosting literature. We illustrate through examples the regularized and asymptotic behavior of the solutions to the classification problem with both loss criteria.

1 Introduction and outline

Boosting is a method for iteratively building an additive model

$$F(\mathbf{x}) = \sum_j \alpha_j h_j(\mathbf{x}), \tag{1}$$

where $h_j \in \mathcal{H}$ and $\mathcal{H}$ is a large (but we will assume finite) family of candidate predictors or "weak learners". Since the introduction of the boosting idea by Freund and Schapire [5], with their famous AdaBoost algorithm, the "mechanism" which makes it work has been presented and analyzed from several perspectives.

In [7], Friedman et al. develop a statistical perspective, which ultimately leads to viewing AdaBoost as a gradient-based incremental search for a good additive model, using the exponential loss function $C(y, F) = \exp(-yF)$, where $y \in \{-1, 1\}$. The gradient boosting [6] and anyboost [9] generic algorithms have used this approach to generalize the boosting idea to wider families of problems and loss functions. In particular, the logistic log-likelihood loss $C(y, F) = \log(1 + \exp(-yF))$ is a more natural loss for classification and is more robust to outliers and misspecified data.

A different analysis of boosting, originating in the Machine Learning community, concentrates on the effect of boosting on the margins $y_i F(\mathbf{x}_i)$. For example, [12] uses margin-

based arguments to prove convergence of boosting to perfect classification performance on the training data under general conditions, and to derive bounds on the generalization error.

In this paper we combine the two approaches, to conclude that gradient-based boosting can be described, in the separable case, as a margin maximizing process.

We consider the problem of minimizing the exponential and logistic loss functions over the training data, with an L_1 bound on the model coefficients:

$$\hat{\beta}(c) = \arg\min_{\|\beta\|_1 \leq c} \sum_i C(y_i, \beta' h(\mathbf{x}_i)). \tag{2}$$

Hastie et al. ([8], chapter 10) have conjectured that gradient-based ϵ-boosting (i.e. $\alpha_j = \epsilon$, $\forall j$ in (1)) , with ϵ infinitesimal, exactly follows the penalized path $\hat{\beta}(c)$ as a function of c. Efron et al. [4] recently proved this result for the case where C is squared error loss, under some mild conditions on the optimal path. Figure 1 illustrates this equivalence between ϵ-boosting and the optimal solution of (2). In this paper we generalize this result to general convex loss functions and conclude that boosting can be viewed as an approximate incremental method for following this regularized path.

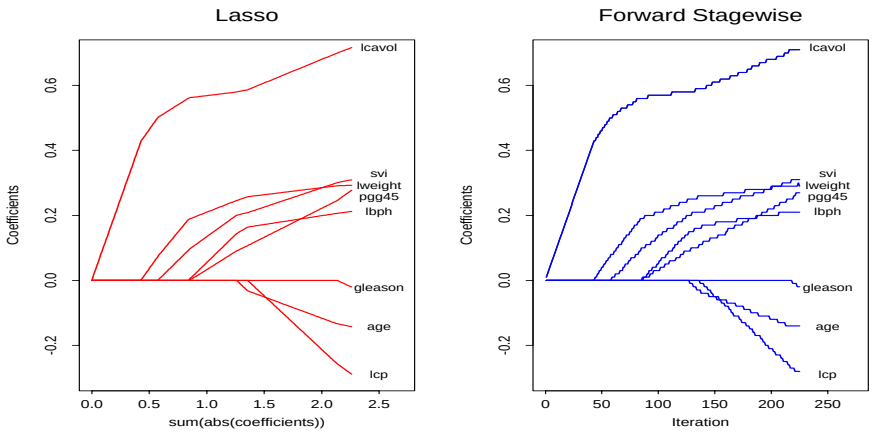


Figure 1: Exact coefficient paths(left) for L_1-constrained squared error regression and "boosting" coefficient paths (right) on the data from a prostate cancer study

We then prove that in the separable case, for both the exponential and logistic log-likelihood loss functions, $\hat{\beta}(c)/c$ converges as $c \to \infty$ to an "optimal" separating hyper-plane $\hat{\beta}$ described by:

$$\hat{\beta} = \arg\max_{\|\beta\|_1 = 1} \min_i y_i \beta' h(\mathbf{x}_i) \tag{3}$$

In other words, $\hat{\beta}$ *maximizes the minimal margin* among all vectors with L_1-norm equal to 1. This result generalizes easily to other L_p-norm constraints. For example, if $p = 2$, then $\hat{\beta}$ describes the optimal separating hyper-plane in the Euclidean sense, i.e. the same one that a non-regularized support vector machine would find.

Combining our two main results, we see that boosting is a gradient-based incremental search, roughly following the path of L_1-constrained optimal solutions, and converging, in the separable case, to a "margin maximizer" in the L_1 sense. Note that boosting with a large dictionary $\mathcal{H}$ practically guarantees that the data will be separable (except for pathologies), hence separability is a very mild assumption here.

As in the case of support vector machines, the asymptotic separating hyper-plane is usually of theoretical interest only, since it typically represents a highly overfitted model. Thus, we

would want to choose a good regularized model. Boosting gives us a natural method for doing that, by "stopping early" in the boosting process.

1.1 Related work

Schapire et al. [12] have identified the normalized margins as distance from an L_1-normed separating hyper-plane. Their results relate the boosting iterations' success to the minimal margin of the combined model. Raetsch et al. [11] take this further using an asymptotic analysis of AdaBoost. They prove that the "normalized" minimal margin, $\min_i y_i \sum_j \alpha_j h_j(\mathbf{x}_i) / \sum_j |\alpha_j|$, is asymptotically equal for both classes. In other words, they prove that the asymptotic separating hyperplane is equally far away from the two closest points on either side. This is a property of the margin maximizing separating hyperplane as we define it. Both papers also illustrate the margin maximizing effects of AdaBoost through experimentation. However, they both stop short of proving the optimality of the hyperplane.

The convergence of boosting to an "optimal" solution has been analyzed in several papers. Raetsch et al. ([10]) and Collins et al ([2]) give results and bounds on the convergence of training-set loss, $\sum_i C(y_i, \sum_j \alpha_j h_j(\mathbf{x}_i))$, to its minimum. However, in the separable case convergence of the loss to 0 is inherently different from convergence of the linear separator to the optimal separator. Any solution which separates the two classes perfectly can drive the exponential (or log-likelihood) loss to 0, simply by scaling coefficients up linearly.

2 Boosting and its interpretations

Generic gradient-based boosting algorithms such as [6, 9] attempt to find a good linear combination of the members of some dictionary of basis functions to optimize a given loss function over a sample. This is done by searching, at each iteration, for the basis function which gives the "steepest descent" in the loss, and changing its coefficient accordingly. Assume we have data $\{\mathbf{x}_i, y_i\}_{i=1}^n$, a loss (or cost) function $C(y, F)$, and a set of basis functions $\{h_j(\mathbf{x})\}$. Then all of these algorithms follow the same essential steps:

1. Set $\beta_0 = 0$.
2. For t = 1: T,
   (a) Let $F_i = \beta'_{t-1} h(\mathbf{x_i}),\ i = 1, \ldots, n$ (the current fit).
   (b) Set $w_i = \frac{\partial C(y_i, F_i)}{\partial F_i},\ i = 1, \ldots, n$.
   (c) Identify j which maximizes $|\sum_i w_i h_j(\mathbf{x_i})|$.
   (d) Set $\beta_{t,j} = \beta_{t-1,j} - \alpha_t \cdot sign(\sum_i w_i h_j(\mathbf{x_i}))$ and $\beta_{t,k} = \beta_{t-1,k}, k \neq j$.

where β_t is the "current" coefficient vector and $\alpha_t > 0$ is the current step size. Notice that $\sum_i w_i h_j(\mathbf{x_i}) = \frac{\partial \sum_i C(y_i, F_i)}{\partial \beta_j}$.

This algorithm can be interpreted as choosing a first order optimal coefficient to increment at each step. Implementation details include the dictionary $\mathcal{H} = \{h_j\}$ of "weak learners", the loss function $C(y, F)$, the method of searching for the optimal j and the way in which α_t is determined. For example, the original AdaBoost algorithm uses this scheme with the exponential loss $C(y, F) = \exp(-yF)$, and an implicit line search to find the best α_t once a "direction" j has been chosen (see [8], chapter 10 and [9] for details). The dictionary used by AdaBoost in this formulation would be a set of candidate classifiers, i.e. $h_j(\mathbf{x_i}) \in \{-1, +1\}$.

Loss functions

The two most commonly used loss functions when boosting classification models are the exponential loss function and the logistic log-likelihood loss function:

$$\begin{aligned} Exponential: &\quad C_e(y,F) = \exp(-yF) \\ Logistic: &\quad C_l(y,F) = log(1+\exp(-yF)) \end{aligned}$$

These two loss functions bear some important similarities to each other:

1. As [7] shows, the population minimizer of expected loss at point $\mathbf{x}$ is similar for both loss functions and is given by: $\hat{F}(\mathbf{x}) = c \cdot log[P(y=1|\mathbf{x})/P(y=-1|\mathbf{x})]$, where $c_e = 1/2$ for exponential loss and $c_l = 1$ for logistic loss.
2. More importantly for our purpose, we have the following simple proposition:

   (4) $$yF \geq 0 \Rightarrow C_e(y,F) \leq C_l(y,F) \leq \log(2) C_e(y,F)$$

   In other words, the two losses become similar if the margins are positive, and both behave like exponentials.

For negative margins the behaviors of C_e and C_l are very different, as [7] have noted. In particular, C_l is more robust against outliers and misspecified data.

Line-search boosting vs. ϵ-boosting

As mentioned above, AdaBoost determines α_t using a line search. In our notation for the boosting algorithm this would be:

$$\alpha_t = |\arg\min_{\alpha} \sum_i C(y_i, F_i + \alpha h_j(\mathbf{x_i}))|$$

The alternative approach, suggested by Friedman [6, 8], is to "shrink" all α_t to a single small value ϵ. This may slow down learning considerably (depending on how small ϵ is), but is attractive theoretically, since the first-order theory underlying gradient boosting implies that the weak learner chosen is the best increment only "locally". In practice, this approach performs well (see [6]). For our purposes, we will mostly assume ϵ is infinitesimally small, so the theoretical boosting algorithm which results is the "limit" of a series of boosting algorithms with shrinking ϵ.

In regression terminology, the line-search version is equivalent to forward stage-wise modeling, infamous in the statistics literature for being too greedy and highly unstable (See [6]). This is intuitively obvious, since by increasing the coefficient until it "saturates" we are destroying "signal" which may help us select other good predictors.

Boosting as incremental L_1-constrained fitting

The way boosting works gives us a natural bound on the L_1 norm of the coefficient vector after t steps: $\|\beta_t\|_1 \leq \sum_{i \leq t} |\alpha_i|$. With ϵ-boosting this is especially useful, since we get $\|\beta_t\|_1 \leq t\epsilon$, so the bound is increasing linearly in boosting iterations. Identification of a single coefficient for modification (step 2(c) of the generic algorithm) can be interpreted as an L_1-optimal choice (as opposed to moving in the direction of the gradient, which is an L_2-optimal choice). This raises the obvious comparison to the constrained optimization problem:

(5) $$\hat{\beta}(c) = \arg\min_{\|\beta\|_1 \leq c} \sum_i C(y_i, \beta' h(\mathbf{x_i})).$$

[8] have shown experimentally (in chapter 10) that for small ϵ and a small number of basis functions, the coefficient paths generated by boosting tend to coincide with the optimal solution $\hat{\beta}(c)$ for the whole range $c \geq 0$. They conjectured that for infinitesimal ϵ, this equivalence holds for any convex $C(y, F)$ if the optimal coefficient paths are monotone in the constraint, i.e. if $\forall j, \quad |\hat{\beta}(c)_j|$ is non-decreasing in c.

Efron et al. [4] recently proved this conjecture for squared error loss regression, i.e. for $C(y, F) = (y - F)^2$. Their proof requires only slight modifications to apply to any "nice" convex loss. Figure 1 illustrates this equivalence for squared error loss. Figure 2 shows a similar result, but this time for logistic log-likelihood loss, C_l. We used the "spam" dataset, taken from the UCI repository [1]. We chose only 5 predictors of the 57 to make the plots more interpretable and the computations more accommodating. We see that there is a prefect equivalence between the exact and stage-wise versions in this case, since the paths are fully monotone. In the context of "real-life" boosting, where the number of

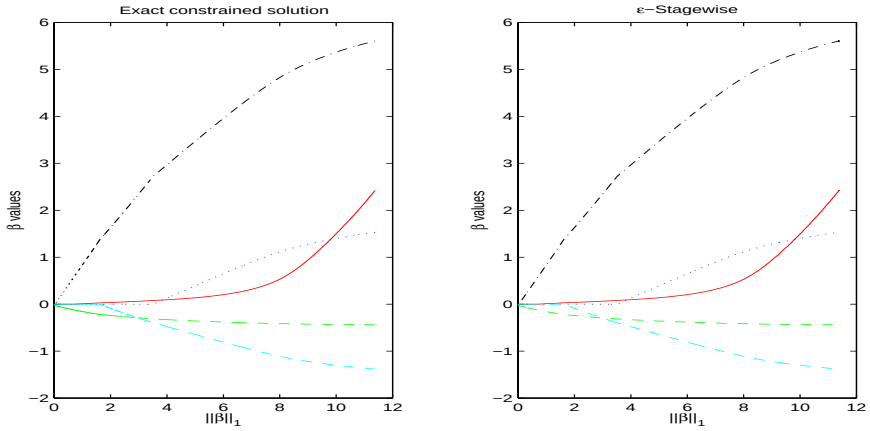


Figure 2: Exact coefficient paths(left) for L_1-constrained logistic regression and boosting coefficient paths (right) on five variables from the "spam" dataset

basis functions is usually very large, and making ϵ small enough for the theory to apply would require running the algorithm forever, these results should not be considered directly applicable. Instead, they should be taken as an intuitive indication that boosting - especially the ϵ version - is, indeed, approximating optimal solutions to the constrained problems it encounters along the way.

3 L_p-constrained classification loss functions

Let us now concentrate on C_e and C_l, the two "classification" losses defined above, and the solution paths of their L_p constrained versions:

$$\hat{\beta}(c) = \arg \min_{\|\beta\|_p \leq c} \sum_i C(y_i, \beta' h(\mathbf{x_i})) \tag{6}$$

where C is either C_e or C_l.

Theorem 1 *Assume the data is separable, i.e.* $\exists \beta \; s.t. \forall i, \; y_i \beta' h(\mathbf{x_i}) > 0$.
Then for both C_e *and* C_l, $\hat{\beta}_c$ *converges to the* L_p*-margin-maximizing separating hyperplane in the following sense:*

$$\lim_{c \to \infty} \frac{\hat{\beta}(c)}{c} = \arg \max_{\|\beta\|_p = 1} \min_i y_i \beta' h(\mathbf{x_i}) \tag{7}$$

Main ideas of proof:
This outline applies for both C_e and C_l, given the property in (4). Let us take two separating candidates, β_1 and β_2 such that $\|\beta_1\|_p = \|\beta_2\|_p = 1$. Assume that β_1 separates better, i.e.: $\min_i y_i \beta_1' h(\mathbf{x_i}) > \min_i y_i \beta_2' h(\mathbf{x_i}) > 0$.
Then from the exponential nature of C we get that:

$$\begin{aligned} \exists c \; s.t. \; \forall d > c, \;\; \sum_i C(y_i, d\beta_1' h(\mathbf{x_i})) &\leq n \exp(d \cdot \min_i y_i \beta_1' h(\mathbf{x_i})) \\ &< \exp(d \cdot \min_i y_i \beta_2' h(\mathbf{x_i})) \\ &\leq \sum_i C(y_i, d\beta_2' h(\mathbf{x_i})) \end{aligned}$$

Thus $c\beta_1$ will eventually becomes superior to $c\beta_2$.
This gives us an order on the p-norm-1 β's, and implies that the asymptotic solution, when normalized, will give us a margin-maximizing separating hyper-plane. □

The margin maximizing separating hyper-plane may not be unique. This theorem can easily be generalized to characterize a unique solution by defining tie-breakers: if the minimal margin is the same, then the second minimal margin determines which model separates better, and so on.

Relation to boosting

Combined with our results from section 2, theorem 1 indicates that the normalized boosting path $\frac{\beta_t}{\sum_{u \leq t} \alpha_u}$ with either C_e or C_l "approximately" converges to a separating hyper-plane $\hat{\beta}$, which attains:

$$\max_{\|\beta\|_1 = 1} \min_i y_i \beta' h(\mathbf{x_i}) = \max_{\|\beta\|_1 = 1} \min_i y_i d_i \|\beta\|_2, \tag{8}$$

where d_i is the Euclidean distance from the training point i to the separating hyperplane. In other words, it maximizes Euclidean distance scaled by an L_2 norm. This implies that the *asymptotic* boosting solution will tend to be sparse in representation, due to the fact that for fixed L_1 norm, the L_2 norm of vectors that have many 0 entries will generally be larger. We conjecture that this asymptotic solution $\hat{\beta} = \lim_{c \to \infty} \hat{\beta}_c / c$, will have *at most* n (the number of observations) non-zero coefficients. This in fact holds for the case of squared error loss, where there exists a finite optimal solution $\hat{\beta}$ (see Efron et al. [4]).

We can also use this theorem to formulate L_p-boosting algorithms, which will approximate the corresponding L_p-margin maximizing separating hyper-planes. Of particular interest is the L_2 case, since our theorem implies that L_2-constrained fitting using C_l or C_e will build a regularized path to the optimal separating hyperplane in the Euclidean (or SVM) sense. The L_1-boosting algorithm as we describe it can in fact be changed quite easily to an L_2-boosting algorithm, by modifying step 2(c) only. Detailed discussion is deferred to a future paper.

4 Regularized behavior of the loss functions

We have shown that boosting with C_e and C_l and support vector machines share the same property of convergence to an "optimal" separator as the regularizer vanishes (or equivalently as the constraint goes to ∞). However, even where it is practical, it is rarely wise to use the non-regularized solutions as prediction models, due to their obvious tendency to be highly overfitted. So the way in which the various criteria and regularizers affect the models generated along the regularized path is a critical issue in determining their practical

advantages and disadvantages. Using our view of boosting as approximate L_1 constrained fitting, it justifies looking at the models along the path as regularized versions of the optimal separator, and thus early stopping in boosting is essentially a choice of a regularization parameter ([6] has previously pointed out this fact).

Our results from section 3 indicate that if the data are separable and we let boosting run forever, we will approach the same "optimal" separator for both C_e and C_l. However if we stop early - or if the data is not separable - the behavior of the two loss functions may differ significantly, since C_e weighs negative margins exponentially, while C_l is approximately linear in the margin for large negative margins (see [7] for detailed discussion). Consequently, we can expect C_e to concentrate more on the "hard" training data in the non-separable case, and to approach the optimal separator more quickly (hence, in some sense, regularize less) in the separable case. Figure 3 illustrates the behavior of ϵ-boosting

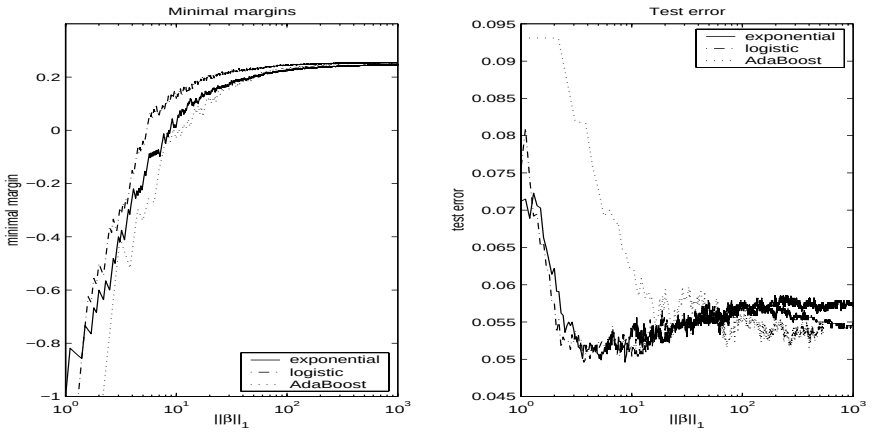


Figure 3: Behavior of boosting with the two loss functions on spam dataset

with both loss functions, as well as that of AdaBoost, on the spam dataset (57 predictors, binary response). We used 10 node trees and $\epsilon = 0.1$. The left plot shows the minimal margin as a function of the L_1 norm of the coefficient vector $\|\beta\|_1$. Logistic loss creates a bigger minimal margin initialy, but the minimal margins for both loss functions are converging asymptotically. AdaBoost initialy lags behind but catches up nicely and reaches the same minimal margin asymptotically. The right plot shows the test error as the iterations proceed, illustrating that both ϵ-methods indeed seem to overfit eventually, even as their "separation" (minimal margin) is still improving. AdaBoost did not significantly overfit in the 1000 iterations it was allowed to run, but it obviously would have if it were allowed to run on.

5 Discussion and future work

This paper introduces a new view of two-class boosting. We hope our results will help in understanding how and why boosting works. For example, so far most of the discussion in the literature on why boosting works well (see [12]) has concentrated on the properties of margins. But according to our view, this may also be due to the implicit use of L_1 regularization by boosting algorithms, and it is well known that regularization usually gives better generalization error. It also comes natural why the ϵ-boosting performs better than AdaBoost [6]: the former follows more closely to the L_1 constraint path. The superiority of L_1 constraint over L_2 constraint (see [3]) may also help explain why boosting works better than the support vector machine [8].

Another important issue is overfitting in the noisy data case. To deal with overfitting, Ratsch et al. [11] propose several regularization methods and generalizations of the original AdaBoost algorithm to achieve a *soft margin* by introducing slack variables. Our results indicate that the models along the boosting path can be regarded as L_1 regularized versions of the optimal separator, hence regularization can be done more directly and naturally by stopping the boosting iterations early. It is essentially a choice of the constraint parameter c.

Many questions also arise from our view of boosting. Among the issues to be considered:

- Is there a similar "separator" view of multi-class boosting? We have some tentative results to indicate that this might be the case if the boosting problem is formulated properly.
- Can the constrained optimization view of boosting help in generating generalization error bounds for boosting that would be more practical than the current existing ones?

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